



National University of Computer & Emerging Sciences, Karachi
Department of Computer Science
Task # 1



Course Code:	Course Name: Theory of Automata
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Write down languages of following set Given alphabet set {a,b}.

$$L1 = \{a^n b^m : (n + m) \text{ is even}\}$$

$$L2 = \{a^n b^m, n \geq 4, m \leq 3\}$$

$$L3 = \{a^n b^m, n < 4, m \leq 4\}$$

$$L4 = \{a^n b^m : n \geq 1, m \geq 1, nm \geq 3\}$$

$$L5 = \{ab^n w : n \geq 3, w \in \{a, b\}^+ \}$$

$$L6 = \{vwv : v, w \in \{a, b\}^*, |v| = 2\}$$

L7 = having exactly one pair of consecutive zeros.

L8 = having exactly one a.

L9 = strings containing no more than 3 a's

L10 = all strings that contain at least one occurrence of each symbol in alphabet

L11 = all strings ending in 0, 1.

L12 = all string not ending in 0, 1

L13 = All strings containing even number of zeros.

L14 = all string having at least two occurrences of substring 00.

L15 = all strings not containing 101.

$$L16 = \{w : |w| \bmod 3 = 0\}$$

$$L17 = \{w : n_a(w) \bmod 3 = 0\}$$

L18 = The language of all strings containing exactly two a's.

L19 = The language of all strings containing at least two a's.

L20 = The language of all strings that do not end with ab.

L21 = The language of all strings that begin or end with aa or bb.

L22 = The language of all strings not containing the substring aa.

L23 = The language of all strings in which the number of a's is even.

L24 = The language of all strings in which both the number of a's and the number of b's are even.

L25 = The language of all strings containing no more than one occurrence of the string aa. (The aaa string contains two occurrences of aa.)

L26 = The language of all strings in which every a (if there are any) is followed immediately by bb.

L27 = The language of all strings containing both bb and aa as substrings.

L28 = The language of all strings containing both aba and bab as substring.

Good Luck